Next, we will have a quick look at alerting with Grafana, enabling you to manage alerts and notifications beyond Prometheus.

7.2.2 Using Grafana for alerting

We discussed Grafana in chapter 6, and now we will have a brief look at its alerting capabilities. Up until version 8, Grafana supported a dashboard-centric alerting model. This means an alert was confined to a dashboard, making it hard to manage notification policies. For example, answering the question, "What sends alarms via PagerDuty to a certain destination?" would require manually checking all dashboards. We will focus on unified alerting (http://mng.bz/GyvM), available from Grafana version 9 and beyond.

First off, why would you want to use Grafana for alerting? In general, your use case covers multiple backends (chapter 5). That is, imagine you are using Prometheus to store metrics and OpenSearch to store logs and traces and want to define alerts across these backends in a central place.

With unified alerting, Grafana now supports these use cases with the following:

- There is support for a range of data sources, including Amazon CloudWatch, Azure Monitor, Google Cloud Monitoring, InfluxDB, Jaeger, and Prometheus.
- You can organize alerts by group (similar to what we've seen in Prometheus).
- There is support for different Alertmanagers, including an integration with the Prometheus one.
- For notifications (http://mng.bz/zX7w), you use so-called contact points, supporting a wide array of types (https://grafana.com/docs/grafana/v9.5/alerting/fundamentals/contact-points/), including Discord, Email, Kafka, PagerDuty, Slack, Telegram, and WebHook.
- You can suppress notifications using silences. Note that the alert rule still gets evaluated by Grafana, causing potential costs for calling the backend, but the notification does not get created.

Now, let's move on to alerting solutions from cloud providers.

7.2.3 Cloud providers

Every major cloud provider offers an alerting solution, typically deeply integrated with its offerings. These offerings are a great baseline you should always consider, since they cover the cloud provider's APIs. These include

- Amazon CloudWatch alarms (http://mng.bz/0Kjp), which uses Amazon SNS to send email or SMS notifications
- Azure Monitor Alerts (http://mng.bz/KeyP), which supports email, SMS, and push notifications
- Google cloud alerting (https://cloud.google.com/monitoring/alerts/), supporting a range of notification types, from email to Pub/Sub